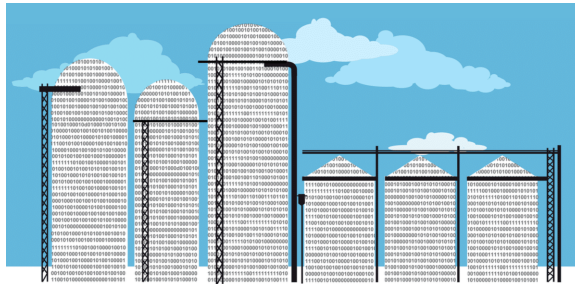




SEMANTIC WEB AND LINKED DATA IN DH

XML	Relational DB	RDF
Hierarchical (tree of elements)	Flat tables (rows and columns)	Graph
Document centric	Data centric	Data centric
Semi-structured data	Structured data	Structured data

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from data silos to web of data
from extension of indices to linked data

APPLICATION OF SEMANTIC WEB TECHNOLOGIES TO DH PROJECTS

TWO LEVELS:

1. Extension of indices (Vogeler 2019)

- Identify entities by linking to **authority files and controlled vocabularies**. This can be done with any technology (XML, relational DB, etc.)

2. Linked data

- Databases of **RDF statements**
- **Persistent URIs** and **existing ontologies** for organising information
- Expose data through a **SPARQL endpoint**

LEVEL 1. EXTENSION OF INDICES

SOME AUTHORITY FILES AND CONTROLLED VOCABULARIES

- [VIAF](#)
- [GeoNames](#)
- Getty Vocabularies:
 - Art & Architecture Thesaurus (AAT)
 - Getty Thesaurus of Geographic Names (TGN)
 - Union List of Artist Names (ULAN)
 - Cultural Objects Name Authority (CONA)
 - Getty Iconography Authority (IA)
- [Iconclass](#)
- Trismegistos
- [Schema.org](#)

LEVEL 2. LINKED DATA

SEMANTIC WEB or WEB OF DATA or WEB 3.0

- **URIs** to identify things (not only documents)

Examples:

Dahlia (plan): <https://www.wikidata.org/wiki/Q130918>

Dahlia de la Cerda: https://dbpedia.org/page/Dahlia_de_la_Cerda

- **Ontologies** to define common vocabularies

FOAF (Friend of a Friend): <http://xmlns.com/foaf/spec/>, CIDOC-CRM: <http://www.cidoc-crm.org/cidoc-crm/>

- **RDF** to express data as triples

`<https://dbpedia.org/page/Dahlia\_de\_la\_Cerda> foaf:birthday "1985-03-09"^^xsd:date ..`

- **SPARQL** to query data

- Linking data across different sources → **Linked Data**

WHY LINKED DATA? SOME EXAMPLES

- **Breaks data silos.** Connect data from different sources.

Examples:

Library records linked to author biographies and book reviews from other sites.

- **Machine-Readable and interoperable. Richer queries and discoveries.** Data is published in standard formats (RDF, URIs) so computers can understand and use it automatically.

A search engine combining data about people, places, and events from diverse websites.

Find all researchers working on climate change by querying scholarly and environmental datasets.

- **Supports data reuse and updating.** Data can be published once and reused by many applications.

Open government data sets reused by civic apps for transparency and services.

HISTORY TIMELINE (1990S TO 2010S)

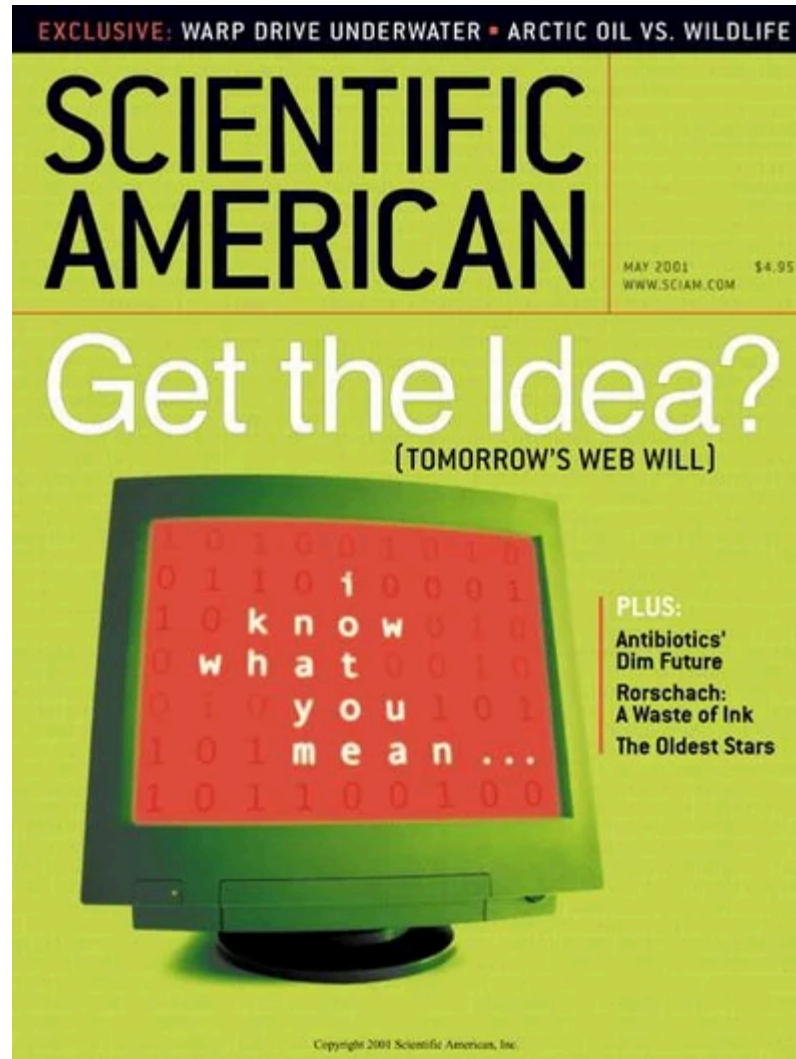
Year	Milestone
1994	Early visions of the Semantic Web
1999	RDF formal W3C Recommendation
2001	Scientific American article formalizes vision
2004	OWL becomes W3C standard
2010s	Growth of Linked Open Data and Schema.org

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What is the W3C?

[World Wide Web Consortium](#) (W3C), international community developing web standards like HTML, CSS, XML, RDF, OWL, SPARQL.



Scientific American, May 2001

"The Semantic Web"

Tim Berners-Lee, James Hendler and Ora Lassila



Introduction to the Semantic Web

Tim Berners-Lee



[Decentralized Information Group](#)

MIT Computer Science and Artificial Intelligence Laboratory

"For example in biopax" (slide 16)

EXAMPLES OF CURRENT USAGE

- An RDF dataset is a **Knowledge Graph**, sometimes abbreviated as KG (but not all KGs are RDF datasets, other technologies can be used).

Examples:

[Google Knowledge Graph](#)

- Semantic web technologies power **search engines**.

[Schema.org](#) has been adopted by web browsers.

The Google KG is used to produce [Knowledge Panels](#).

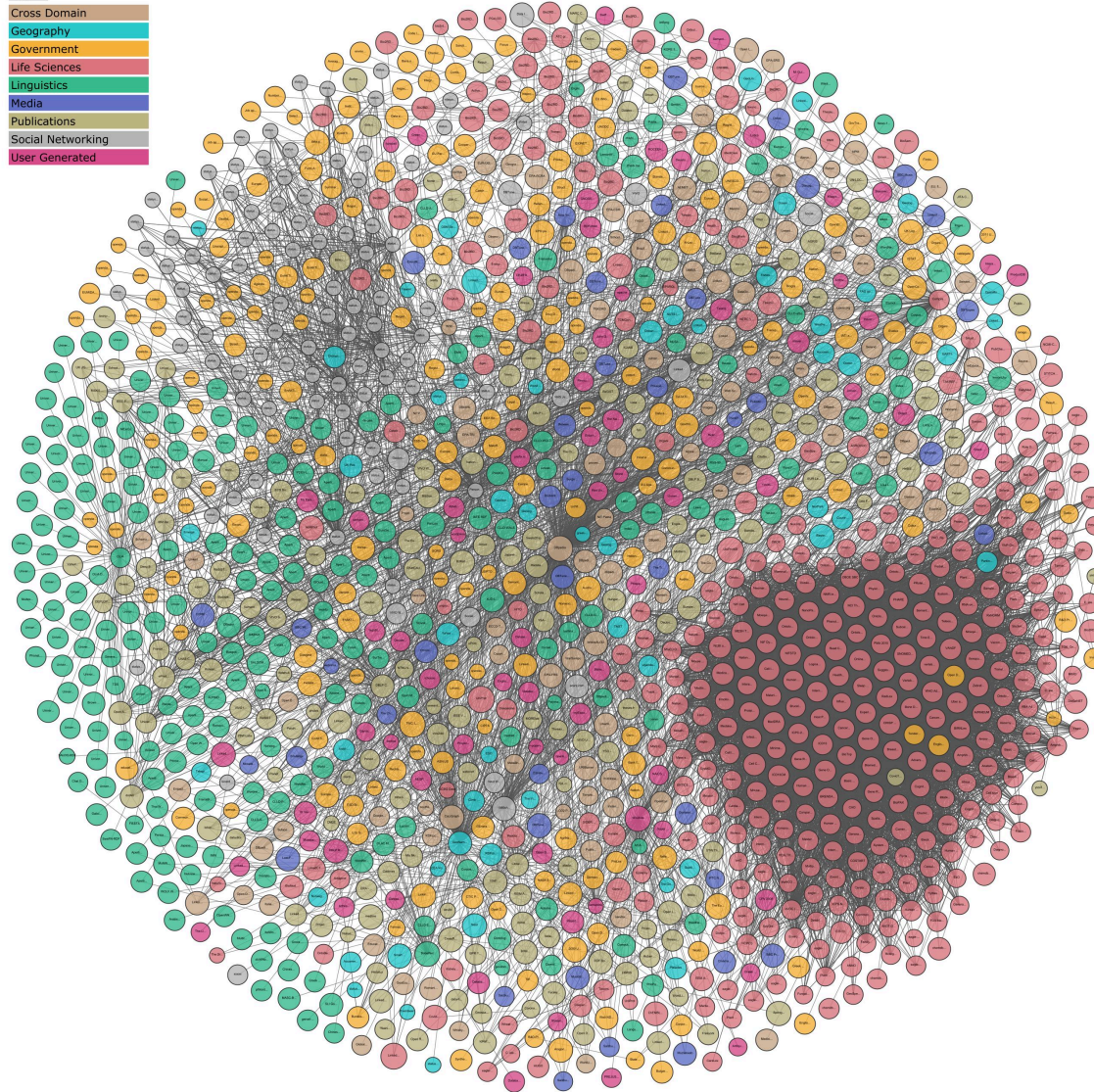
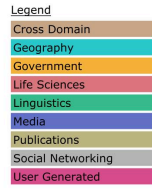
- Semantic web technologies are used in **healthcare, bioinformatics, smart utilities, cultural heritage, government, and more.**

[Swiss Institute of Bioinformatics data resources](#)

[Swiss Food Knowledge Graph](#)

[Swiss Open Government data](#)

The Linked Open Data Cloud



ONTOLOGIES COMMONLY USED IN DH

- [CIDOC CRM](#): Conceptual Reference Model for cultural heritage information
- [LRMoo](#): Library Reference Model for bibliographic information (previously FRBRoo)
- [Dublin Core](#): Metadata terms for resource description
- [SPAR Ontologies](#): Ontologies for citations, bibliographic references, and more
- [FOAF](#): Friend of a Friend vocabulary for describing people and their relationships
- [PROV-O](#): Provenance Ontology for representing provenance information
- [RIC-O](#): Records in Context Ontology for bibliographic records

SOME DH RELATED PUBLIC SPARQL ENDPOINTS

- [Europeana](#): digital cultural heritage platform.
- [Mapping Manuscript Migrations](#): pre-modern manuscripts and their movements.
- [MiMo TextBase](#): French Enlightenment novel (1750-1800).
- [LiLa \(Linking Latin\)](#): linguistic resources for latin.
- [German National Library](#): Linked Data service.
- [BnF Data](#): open data of the French National Library.